



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Computing & Applied Sciences Department CS&IT

Semester 2nd Degree Program BS (CS) & (IT) Batch 2021F

Course Code: CS-127T Course Title: Object Oriented Programming Credit Hours: 3+0

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.		Name		'A' Copy Serial No.	
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Q.No.1 **5 + 5 Marks**

- Write a C# program that takes an integer array {10, 20, 50, 110, 90, 30, 100, 40, 80, 70} and asks user to enter a number to be searched in the given array. Use linear search algorithm.
- An automated teller machine (ATM) is a banking subsystem that provides bank customers with access to financial transactions in a public space without the need for a cashier, clerk, or bank teller. Customer uses bank ATM to Check Balances of his/her bank accounts, Deposit Funds, Withdraw Cash and/or Transfer Funds. ATM Technician provides Maintenance and Repairs. All these use cases also involve Bank actor whether it is related to customer transactions or to the ATM servicing. Draw UML Use Case and Sequence diagram of the given system.

Q.No.2 **3 + 7 Marks**

- Write down the missing code:

```
interface Itest
{
    void display(string b);
}
interface Itest2 : Itest
{
    string info();
}
```



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```

abstract class Animal
{
    public abstract void animalSound();
    public void sleep()
    {
        Console.WriteLine("Animal is sleeping");}
}
class Cat : Animal, Itest2
{
    *INSERT MISSING CODE HERE*
}
class Program
{
    static void Main(string[] args)
    {
        Cat an = new Cat();
        string x = an.info();
        an.display(x);
        an.animalSound();
        an.sleep();
    }
}

```

- b) Write a C# program to show the concept and difference of System/Built-in exceptions and application/custom exceptions. Write down pros and cons of custom exceptions

Q.No.3

5 + 5 Marks

- a) Write down code snippets to accomplish the following tasks using sorting/searching built-in C# algorithms. Taking array string [] names = {"Olive", "Patty", "Richard", "Ned", "Mindy", "Pakistan", "Karachi"};
- 1) Apply C# binary search algorithm on the given array.
 - 2) Sort the given string array in Alphabetical (Dictionary type) order by using C# sorting algorithm.
 - 3) Sort given string array in descending order by using C# reverse & sorting algorithm.



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```
{  
    static int a;  
    public abstract void info(int rollno);  
  
    public void print()  
    { a = 15;  
      Console.WriteLine("ABC"); }  
}  
class Test : Sample, ITest  
{  
    static int b(int x)  
    { Console.WriteLine(a);  
      return ++x; }  
}
```

- b) Assume the method display() for overloading and displayinfo() for overriding. First display shows the age of a person and displayinfo shows the age of a person with message younger or older by checking age of a person. Program must show the concepts of overloading and overriding.

Best of Luck ☺